

The Synchronization-Closure Hypothesis

A falsifiable programme from a locally invariant spin core to endogenous geometry, matter candidates, and reciprocal coupling

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Hypothesis and research-programme paper — 7 September 2026

Abstract

This paper formulates a research hypothesis rather than a physical theory. Its starting point is a locally invariant Lorentzian spin core: a fixed Minkowski vector model together with a two-to-one spin cover whose local algebraic structure is taken to be identical in every fibre. The proposed new ingredient is not a preferred frame, a prescribed gravitational field, or a relabelling of standard curvature as “synchronization”. The hypothesis is that the relations by which many locally invariant fibres are compared admit a single, gauge-covariant, self-consistency law. The same law is required to be reciprocal: perturbing the local/relative state must induce a compatible change of relational geometry, while perturbing the relational geometry must alter the transport/evolution of a test state without changing the fibrewise Minkowski core. In a many-fibre system, the background is therefore endogenous rather than externally fixed.

The mathematical prototype is developed first on a graph, where edge transports take values in a spin group and loop holonomy measures failure of global synchronizability, and then in the continuum, where a principal spin bundle, a solder form (coframe), and a spin connection produce a spacetime metric, torsion and curvature while the internal Minkowski metric remains fixed. These geometric structures are standard; the proposed novelty is the existence of a deeper closure law, derivable from the reconstructed spin/Clifford dependency chain rather than postulated as the Einstein-Hilbert, Yang-Mills, geodesic, lapse, redshift, or mass equations.

A precise hierarchy of tests is specified. The flat sector must reproduce special-relativistic kinematics with no preferred physical frame. The collective sector must generate geometry self-consistently and recover free-fall and general-relativistic benchmarks without inserting the Einstein equations as primitives. Stable localized solutions, if they exist, become matter candidates; mass is to be defined only after an asymptotic energy-momentum invariant exists, rather than supplied as an elementary source parameter. Propagating relaxation modes are radiation candidates only if their helicity, dispersion, quantization and thermodynamic limits pass the relevant tests. Three generations, confinement, asymptotic freedom, deconfinement and fusion are explicitly treated as later falsification targets, never as input assumptions. The paper also states a scale-provenance requirement: no dimensionful mass or gravitational scale may appear without a traceable origin in state, boundary data, quantization/renormalization, or genuine dynamical scale generation.

Status and object table

Throughout the paper, every claim belongs to one of four classes: F = current formal baseline or already kernel-checked project input; S = established mathematics/physics supported by primary literature; H = new hypothesis proposed here; T = future benchmark or falsification test. No H or T statement is presented as established fact.

Table 1: Objects, meanings, and admissible interpretations.

Object	Mathematical form	Status	Meaning and restriction
Local vector model	(V, η) $(\mathbb{R}^{1,3}, \text{diag}(+, -, -, -))$	$\cong$ F/S	Fibrewise Lorentzian quadratic space. “Invariant” means the same local model is used in every fibre; it does not mean that all fields or states are constant.
Local spin cover	$\rho: \tilde{G} \twoheadrightarrow G \subseteq \text{SO}^+(1, 3)$ with $\ker \rho = \{\pm 1\}$	F/S	Algebraic double cover. In the current project the native pair is $(\text{Spin}_{\text{alg}}, G_{\text{L}})$; the fully topological/global identification remains part of the ongoing programme.
Spin/Clifford core	$\mathcal{K} = (V, \eta, \text{Cl}, \tilde{G}, \rho)$	H	Name for the minimal locally invariant package used by this paper. It is not claimed to be a complete ontology.
Fibre/site	$F_x \simeq \mathcal{K}$	H	One local copy of the invariant core. Many fibres are required for the collective hypothesis.
Relative transport	$U_{xy} \in \tilde{G}$	S/H	Discrete comparison between neighbouring fibres. It is gauge dependent; only gauge-invariant/comparison-invariant information is physical.
Discrete solder data	$\ell_{xy} \in V_x, \ell_{yx} = -\rho(U_{xy})\ell_{xy}$	H	Optional graph-level analogue of a coframe/edge displacement. It is required if the discrete prototype is to encode metric/torsion information; spin transport U alone encodes only frame comparison/holonomy.
Synchronization class	$[U]$ modulo local frame changes	H	Precise replacement for the heuristic word “synchronization”. It denotes relational comparison data, not a scalar clock field.
Synchronization zero	$[U]$ gauge-equivalent to $U_{xy} = 1$ on a simply connected region	H	Neutral relational configuration. It is not an absolute physical frame.
Loop mismatch	conjugacy class $[H_C]$, $H_C = \prod_{e \in C} U_e$	S/H	Gauge-invariant obstruction to trivializing a closed comparison loop. In the continuum its infinitesimal analogue is curvature/holonomy.
Continuum solder form	$e: TM \xrightarrow{\sim} E = P \times_{\rho} V$	S	Identifies tangent vectors with the internal Minkowski bundle; it induces $g = e^*\eta$.
Spin connection	principal connection ω on P	S	A $\mathfrak{g}$ -valued principal connection one-form satisfying equivariance and reproduction of vertical generators; it governs parallel comparison of local spin frames.
Curvature	$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega]$	S	Infinitesimal holonomy; standard differential geometry, not itself a new law of gravity.
Torsion	$T = D_{\omega}e = de + \rho_*(\omega) \wedge e$	S	Failure of the solder form to be covariantly closed. $T = 0$ is a compatibility condition, not a dynamical field equation.
Closure law	$\mathfrak{E}(\Phi, e, \omega) = 0$ or discrete analogue	H	Unknown self-consistency relation sought by the programme. It must not be defined by inserting the Einstein, geodesic, lapse, redshift, Yang-Mills or mass equations.
Matter candidate	stable, localized, finite-charge/finite-energy non-trivial solution modulo gauge	T	“Matter” is earned only after stability and conserved/asymptotic invariants exist.
Radiation candidate	propagating linear/nonlinear perturbation carrying finite flux	T	Not called photon or thermal radiation until helicity, dispersion, quantization and thermodynamics are recovered.
Mass	asymptotic invariant $m^2 = P_a P^a$ when P^a exists	T	Not a primitive source in the mature model. A dimensionful scale must have explicit provenance.
Generation	distinct stable branches with otherwise matching observable quantum labels	T	The number three must emerge or the generation hypothesis fails.

Object	Mathematical form	Status	Meaning and restriction
Confinement	gauge-invariant non-separability benchmark, e.g. nonzero string tension/area-law regime	T	Only meaningful if an appropriate non-Abelian internal gauge sector is first derived.

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1 Methodological discipline and scope

1.1 Writing and claim discipline

The exposition follows the mathematical-writing discipline advocated by Tao: state the principal claim early, distinguish proved statements from motivation, describe limitations openly, and organize by logical dependence rather than chronology [1]. Accordingly, the central proposal is explicitly labelled a hypothesis, and every contact with established relativity, spin geometry, gauge theory or QCD is used as a benchmark rather than as evidence that the hypothesis is true.

1.2 What is not being claimed

This paper does *not* claim that curvature is newly discovered as a synchronization defect, that spin structure by itself implies Einstein gravity, that matter is already derived, that photons are longitudinal modes, that there are necessarily three generations, or that QCD follows from the Lorentz spin group. Local Lorentz frames, spin connections, tetrads/coframes, torsion, curvature and holonomy are established structures dating through Cartan, Fock, Weyl and later gauge-gravity formulations [2, 3, 4, 7, 5]. The prospective novelty is narrower and therefore falsifiable: whether a *single deeper closure principle*, native to the reconstructed spin/Clifford dependency chain, can determine the relational background and localized/propagating excitations reciprocally.

1.3 Cut-DAG viewpoint

A finite model is a cut through a larger causal/dependency graph. Variables entering through the cut are *exogenous relative to the cut*, but need not be fundamental in the hypothetical complete graph. This motivates a strict distinction:

- a *test perturbation* may be imposed as boundary/initial data in order to probe a response;
- a *law parameter* is part of the equations themselves;
- a *derived observable* must not be silently promoted to either of the above.

Thus a relative state perturbation may be injected in a kinematic smoke test, and a background perturbation may initially be imposed in a reciprocal smoke test. The mature many-body model, however, must generate its background self-consistently rather than permanently receiving a lapse, redshift profile or mass parameter from outside.

2 Formal baseline: the locally invariant spin core

2.1 Minimal abstract core

Definition 2.1 (Locally invariant spin core). A *locally invariant spin core* is a tuple

$$\mathcal{K} = (V, \eta, \tilde{G}, G, \rho), \tag{1}$$

where (V, η) is a real four-dimensional Lorentzian vector space of signature $(1, 3)$, $G \leq \text{SO}^+(V, \eta)$, $\tilde{G}$ is a group, and

$$\rho : \tilde{G} \twoheadrightarrow G, \quad \ker \rho = \{\pm 1\} \tag{2}$$

is a surjective two-to-one homomorphism. A field of such cores is *locally invariant* if every fibre is isomorphic to the same tuple $\mathcal{K}$, with the same fibrewise quadratic form and group law.

The current Lean development provides an algebraic instance of (2): an intrinsic spin group, an intrinsic proper future-cone-preserving Lorentz group, a surjective cover map, and exact central kernel $\{\pm 1\}$. This paper uses only that abstract achievement. The global topological spin structure, spin connection and continuum fibre geometry are not assumed to have been completed by the current code.

2.2 Clifford relation and local Minkowski invariance

For a Lorentzian vector space, the Clifford relation is

$$uv + vu = 2\eta(u, v) 1. \quad (3)$$

Matrix gamma relations are representations of (3), not the globally fundamental object. Fock and Weyl already showed in 1929 how spinorial structures can be coupled to curved spacetime through local frames and connections [3, 4]. The hypothesis here therefore does not identify “local Minkowski” with “globally flat”. The local invariant is the fibre model η ; global geometry is allowed to vary through the way fibres are soldered and compared.

Research criterion 2.2 (No local deformation shortcut). Any later geometric or matter sector must preserve the local fibre model up to its allowed local gauge/frame action. A proposed mechanism that obtains gravity merely by replacing the fibre metric η with an arbitrary position-dependent metric has bypassed, rather than explained, the synchronization problem.

3 Discrete relational prototype: many invariant fibres

3.1 Graph of fibres

Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a connected oriented graph. Assign to each vertex $x \in \mathcal{V}$ one copy F_x of the local core. For every oriented edge $x \rightarrow y$, introduce a comparison variable

$$U_{xy} \in \tilde{G}, \quad U_{yx} = U_{xy}^{-1}. \quad (4)$$

No absolute frame is chosen. A local frame change $h_x \in \tilde{G}$ acts by

$$U_{xy} \mapsto U'_{xy} = h_y U_{xy} h_x^{-1}. \quad (5)$$

If a local core state Φ_x transforms in a representation R of $\tilde{G}$, then

$$\Phi_x \mapsto R(h_x) \Phi_x. \quad (6)$$

If the graph is also intended to approximate metric/solder geometry rather than connection holonomy alone, introduce an oriented edge displacement $\ell_{xy} \in V_x$ with

$$\ell_{yx} = -\rho(U_{xy})\ell_{xy}, \quad \ell_{xy} \mapsto \rho(h_x)\ell_{xy}. \quad (7)$$

The pair (U, ℓ) is then the discrete analogue of connection plus solder data. The transport U by itself does not determine a spacetime metric.

Definition 3.1 (Synchronization configuration). A *synchronization configuration* on Γ is a gauge-equivalence class $[U]$ of edge variables under (5). The word synchronization refers only to relational comparison data; it does not introduce a preferred clock or rest frame.

3.2 Loop holonomy and neutral synchronization

For a based closed loop $C = (x_0 x_1 \cdots x_n = x_0)$ define

$$H_C := U_{x_{n-1}x_n} \cdots U_{x_0x_1} \in \tilde{G}. \quad (8)$$

Under (5),

$$H_C \mapsto h_{x_0} H_C h_{x_0}^{-1}. \quad (9)$$

Hence the conjugacy class $[H_C]$ is gauge invariant.

Definition 3.2 (Synchronization zero). On a connected region $R \subset \Gamma$, a synchronization configuration is *neutral* if it is gauge-equivalent to $U_{xy} = 1$ on every edge of R . Equivalently, $H_C = 1$ for every closed loop C in R . If R is the one-skeleton of a simply connected cell complex, it is sufficient to verify this on a generating set of face loops.

Warning 3.3 (No absolute physical frame). The representative $U_{xy} = 1$ is a gauge-fixed neutral configuration, not an absolute inertial frame. A different local frame choice changes the representative while leaving the neutral equivalence class unchanged. Any observable prediction that depends on choosing this representative rather than its class violates the intended relativistic symmetry.

3.3 Why many fibres matter

For one isolated fibre there is no closed comparison loop and therefore no relational holonomy. Nontrivial collective information begins with networks of comparisons. In a non-Abelian group, loop products are order dependent, so a many-fibre configuration cannot generally be reduced to a scalar sum of pairwise mismatches. This is the first precise sense in which “many on a heap” can carry information that no isolated fibre possesses.

Hypothesis 3.4 (Collective relational hypothesis). The physically relevant background variables are functions of the joint relational configuration—connection comparison $[U]$, any required solder data $[\ell]$, and the joint local-core configuration $[\Phi]$ —not a superposition of independently assigned scalar gravitational charges. In particular, collective nontriviality may be encoded in loop/holonomy structure even though every individual fibre retains the same local core.

4 Continuum translation: soldering, connection, curvature and torsion

4.1 Spin bundle and internal Minkowski bundle

Let M be a smooth four-manifold and $P \rightarrow M$ a principal $\tilde{G}$ -bundle. Let

$$E := P \times_{\rho} V \quad (10)$$

be the associated Minkowski vector bundle. The fibre metric η is fixed and ρ -invariant.

The ordinary existence of a spin structure is a global topological problem. For the target double cover

$$\text{Spin}^+(1, 3) \longrightarrow \text{SO}^+(1, 3)$$

used here, one first distinguishes two separate reductions: manifold orientation and time orientation, together reducing the orthonormal-frame structure group to the proper orthochronous Lorentz group. Orientability of TM is detected by $w_1(TM) = 0$; time orientability is a distinct condition and must not be conflated with it. After the relevant oriented Lorentz frame bundle is fixed, the ordinary spin-lift obstruction is $w_2(TM)$. Milnor gives a foundational account of

spin structures and the second Stiefel–Whitney obstruction [9]; Geroch analyzes the Lorentzian spacetime problem and emphasizes that a spinor structure is additional global structure beyond the Lorentz metric [10, 11]. These results are prerequisites for a full continuum implementation; they are not consequences of the synchronization hypothesis.

4.2 Solder form and induced spacetime metric

Definition 4.1 (Solder/coframe field). A solder form is, equivalently for the present purpose, a vector-bundle isomorphism

$$e : TM \xrightarrow{\sim} E. \quad (11)$$

It induces a spacetime metric by pullback:

$$g(X, Y) := \eta(eX, eY), \quad g = e^*\eta. \quad (12)$$

Equation (12) is the exact mathematical expression of “locally Minkowskian fibres, globally variable geometry”: η is unchanged inside every fibre, while the spacetime metric can vary because e varies. Fock and Weyl used local frame/spinorial machinery in general relativity in 1929 [3, 4]; Kibble later derived vierbein and local affine-connection variables from localizing the inhomogeneous Lorentz symmetry [7]. Thus the existence of this mechanism is standard. The new question is whether the present project can *derive the necessary coupling/closure* rather than importing one of these gravitational actions.

4.3 Spin connection, curvature and torsion

Let ω be a principal connection on P , locally represented by a $\mathfrak{g}$ -valued one-form. Its curvature is

$$\Omega = d\omega + \frac{1}{2}[\omega \wedge \omega]. \quad (13)$$

The solder form has covariant exterior derivative

$$T := D_\omega e = de + \rho_*(\omega) \wedge e, \quad (14)$$

which is torsion. Cartan’s affine-connection framework introduced precisely this distinction between curvature and torsion [2]. Ambrose and Singer proved that curvature generates the holonomy algebra, making the continuum connection between loop mismatch and curvature mathematically exact [5].

Warning 4.2 (Geometry is not yet dynamics). The equations (12)–(14) are definitions/kinematics of differential geometry. Neither $T = 0$ nor nonzero Ω explains why a physical background should take one configuration rather than another. Calling Ω a “synchronization defect” adds no physics unless a further law determines how ω and e respond to the local-core configuration.

5 The central new hypothesis: synchronization closure

5.1 Configuration spaces

Let $\mathcal{X}$ denote a configuration space of local-core states Φ (sections of appropriate associated bundles, or discrete vertex states), and let $\mathcal{G}$ denote a configuration space of relational geometry, represented in the continuum by (e, ω) modulo appropriate gauge/frame symmetries. Let $\mathcal{H}$ act on both.

Definition 5.1 (Closure relation). A *synchronization closure relation* is a gauge-covariant constraint

$$\mathfrak{C}(\Phi, e, \omega) = 0 \quad (15)$$

whose zero set is gauge invariant. The corresponding physical solution space is schematically

$$\mathcal{S} := \mathfrak{C}^{-1}(0)/\mathcal{H}. \quad (16)$$

The relation is *endogenous in the background* if (e, ω) are variables solved by (15), not permanently prescribed external functions.

Definition 5.2 (Native and minimal closure). A closure is *native* if every object entering its local law is either already present in the reconstructed dependency spine, forced by the geometric completion of that spine, or explicitly declared boundary/initial data. It is *minimal relative to a stated candidate class* if removing any non-gauge argument or independent coupling from the law destroys at least one required property (gauge covariance, nontrivial collective response, reciprocity, or the neutral sector). “Minimal” is therefore always relative to a declared class of admissible closures and is not asserted absolutely before such a class is classified.

Hypothesis 5.3 (Synchronization-Closure Hypothesis (SCH)). There exists a native closure relation (15) that is minimal in an explicitly classified admissible candidate class and such that:

- SCH1. Local-core invariance.** Every local fibre retains the same Minkowski/spin core up to its gauge/frame action.
- SCH2. Relational geometry.** Global geometry is carried by the soldering/comparison data and their compatibility, not by deforming the local invariant metric itself.
- SCH3. Gauge covariance.** The law is independent of arbitrary local spin-frame choices.
- SCH4. Neutral solution.** There is a locally flat/pure-gauge synchronization-zero sector.
- SCH5. Collective dependence.** The geometric response depends on the joint many-core configuration; no elementary scalar “mass source” is assumed at the fundamental level.
- SCH6. Reciprocity.** The same solution relation couples state and geometry in both directions: admissible state-side perturbations require geometric readjustment, and admissible geometry-side perturbations modify state transport/evolution.
- SCH7. No target smuggling.** Lorentz boost formulae, the geodesic equation, an externally prescribed lapse/redshift profile, Einstein’s field equations, Maxwell/Yang–Mills equations, particle masses, generations, confinement and QCD beta functions are not definitions of $\mathfrak{C}$.

5.2 Precise meaning of bidirectionality

“Bidirectional” is constitutive, not retrocausal. Near a regular solution $z_0 = (\Phi_0, G_0)$, with $G = (e, \omega)$, linearize

$$D\mathfrak{C}_{z_0}(\delta\Phi, \delta G) = C_\Phi \delta\Phi + C_G \delta G. \quad (17)$$

After quotienting gauge zero-modes and imposing admissible boundary conditions, the closure is *two-sided regular* if the solution manifold can locally be solved for the compensating geometric response on the relevant state-driven subspace and for the compensating state response on the relevant geometry-driven subspace. In the simplest invertible linear model,

$$\delta G = -C_G^{-1} C_\Phi \delta\Phi, \quad (18)$$

$$\delta\Phi = -C_\Phi^{-1} C_G \delta G. \quad (19)$$

Equations (18)–(19) are only a local linear template; the full nonlinear theory need not possess globally invertible partial derivatives.

Research criterion 5.4 (Reciprocal-law test). If the state-driven and geometry-driven responses require two independent constitutive laws with unrelated parameters, the central unification hypothesis fails. Different boundary conditions are allowed; different fundamental response laws are not.

5.3 Variational closure as a stronger candidate, not an assumption

A particularly economical realization would be a single gauge-invariant action

$$\mathcal{A}[\Phi, e, \omega] \quad (20)$$

with simultaneous stationarity equations

$$\frac{\delta \mathcal{A}}{\delta \Phi} = 0, \quad \frac{\delta \mathcal{A}}{\delta e} = 0, \quad \frac{\delta \mathcal{A}}{\delta \omega} = 0. \quad (21)$$

This naturally implements the “fixed-resistor” intuition: no side is a passive externally programmed response table; all fields satisfy one common constitutive principle. However, (20) is a *stronger subhypothesis*. The research programme must first ask whether the existing dependency structure selects such a variational principle. It must not simply insert the Einstein–Hilbert, Palatini, Einstein–Cartan or Yang–Mills action. Kibble’s 1961 construction is a primary benchmark showing how vierbein and connection can arise in a gauge-gravity action and how spin density modifies the connection equation [7]; reproducing that structure by assumption would not establish the present hypothesis.

6 Discrete self-regulation prototype without committing to a law

To make the hypothesis computable before continuum geometry is complete, one may study a class of gauge-invariant discrete functionals without treating any member as the final theory. Let p range over elementary loops/plaquettes and let $f : \tilde{G} \rightarrow \mathbb{R}_{\geq 0}$ be a class function with $f(1) = 0$. Let V be a gauge-invariant nearest-neighbour compatibility term. Then

$$\mathcal{E}_{f,V}[U, \Phi] = \sum_p f(H_p) + \sum_{(xy) \in \mathcal{E}} V(\Phi_y, U_{xy} \Phi_x) \quad (22)$$

is gauge invariant under suitable choices of V .

Remark 6.1. Equation (22) is a *search family*, not a proposed law. In this connection-only form it probes reciprocal spin/frame comparison but does not yet encode a full metric background; a geometric prototype must extend it by native terms involving the solder variables ℓ (or their continuum limit e). Varying the relational variables then makes the background respond to the many-core state, while varying Φ makes the state respond to the same background. Wilson’s lattice gauge construction is the canonical primary example showing how gauge-invariant information can be concentrated in edge variables and closed loops [14]. The present programme must not infer QCD from this formal resemblance.

Research criterion 6.2 (Prototype success). A discrete candidate is worth promoting only if it exhibits, without inserting relativistic target equations, all of the following: a gauge-invariant neutral sector, nontrivial collective loop response, two-sided state/geometry response, a controlled continuum limit, and no preferred physical frame.

7 Flat-sector endpoint: special relativity as a terminal comparison

7.1 Neutral geometry and relative group state

A neutral connection class by itself does not guarantee a flat metric unless the solder sector is compatible. Define the *flat neutral sector* locally by trivial connection holonomy together with vanishing torsion (equivalently, in a contractible patch, by a gauge/frame in which the compatible coframe is integrable and the connection is pure gauge). Then the induced metric is locally Minkowskian. Two local frames may nevertheless differ by a constant element $s \in \tilde{G}$. Their observable vector-frame relation is

$$\Lambda = \rho(s) \in G \subseteq \text{SO}^+(1, 3). \quad (23)$$

The identity $s = 1$ is the neutral relative configuration, not a preferred cosmic frame.

For a one-parameter boost subgroup $s(\eta)$, rapidity η is a coordinate on that subgroup. Only at the final comparison stage may one identify the corresponding relative velocity by the usual special-relativistic map $v/c = \tanh \eta$.

Benchmark 7.1 (SRT benchmark). With zero collective curvature/torsion and a state-side relative perturbation, the model must reproduce the Lorentz composition law, invariant light cone/interval, time-dilation/length-contraction relations and relativity of simultaneity from the native group projection. No physical observable may depend on the choice of synchronization-zero representative.

Falsification criterion 7.2 (Preferred-frame falsifier). If the neutral configuration becomes experimentally distinguishable from a gauge-related one, or if the response depends on an absolute velocity rather than relational group data, the proposed route fails its SRT endpoint.

8 Collective geometric endpoint: gravity must be an output

8.1 Endogenous background from many cores

In the mature model a compact or dense many-core configuration is not assigned an external Schwarzschild lapse or a primitive mass charge. Instead it changes the solution (e, ω) of the same closure relation. A distant test core is then placed in this self-consistent background without a direct force term.

Einstein's 1916 formulation makes the metric field dynamical and relates spacetime geometry to matter through field equations [12]; Schwarzschild immediately supplied the exact exterior field of a spherically symmetric source [13]. Those results are benchmarks, not inputs here.

Benchmark 8.1 (Free-fall benchmark). In a regime where internal spin structure of the test object is negligible, the effective trajectory must reduce to metric free fall (or an observationally equivalent law) in the background generated by the other cores. A direct externally inserted "gravitational force" on the test core is not allowed.

Benchmark 8.2 (Macroscopic GR benchmark). The long-distance, low-curvature limit must recover Einstein gravity or a quantitatively viable alternative. At minimum the theory must pass the Newtonian limit, gravitational redshift, light deflection, perihelion advance, Shapiro delay, gravitational-wave propagation/polarizations and equivalence-principle constraints. If Einstein's equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (24)$$

is recovered, (24) must arise as an effective closure equation rather than be inserted into $\mathfrak{C}$.

8.2 Torsion is a live alternative, not a shortcut

Because the natural variables are e and ω , torsion need not be set to zero prematurely. Cartan's geometry and Kibble's gauge-gravity construction show that spin can source torsion-like connection degrees of freedom [2, 7]. The programme must therefore determine whether the deeper closure enforces $T = 0$, permits algebraically determined torsion, or propagates torsion. The answer is a derived branch point with direct experimental consequences.

9 Scale provenance: a mandatory audit for mass and gravity

Research criterion 9.1 (Scale-provenance principle). Every dimensionful output must have a traceable dimensionful origin. A purely algebraic collection of dimensionless groups, topological classes and incidence relations cannot by itself determine a numerical mass, length, time or Newton constant in physical units. Any emergent scale must enter through one or more of:

- (i) boundary/initial data carrying a scale;
- (ii) a dimensionful parameter in a deeper law;
- (iii) a condensate or stable solution whose normalization introduces a scale;
- (iv) quantization/renormalization and dimensional transmutation;
- (v) a conversion scale inherited from a still deeper layer of the full dependency DAG.

The provenance must be recorded, not hidden by a choice of units.

Asymptotic freedom provides an important benchmark for what genuine scale generation can look like: Gross and Wilczek and independently Politzer showed that non-Abelian gauge couplings run to weak coupling at high momentum [15, 16]. In QCD a dimensionful scale is generated through renormalization-group dimensional transmutation once a renormalization condition is supplied. This is an analogy for scale provenance, not evidence that the synchronization closure is QCD.

10 Matter as stable collective structure, not an input source

10.1 Operational definition of a matter candidate

Assume a closure law and a well-defined conserved energy/charge structure have been obtained. Let $(\Phi_\star, e_\star, \omega_\star)$ be a nontrivial solution asymptotic to the neutral vacuum.

Definition 10.1 (Matter candidate). A *matter candidate* is a gauge-equivalence class of solutions that is:

- (i) spatially localized relative to an asymptotic neutral background;
- (ii) dynamically or topologically stable on the timescale of interest;
- (iii) finite in the conserved energy/charge norms supplied by the theory;
- (iv) not removable by a compactly supported gauge transformation;
- (v) reproducible under translations/boosts as appropriate in the asymptotic sector.

Skyrme's 1961 nonlinear field theory is a primary historical example in which particle-like objects arise as finite-energy nonlinear field configurations rather than elementary point sources [8]. This demonstrates mathematical possibility, not the correctness of the present mechanism.

10.2 Mass must be measured after the state exists

If an isolated matter candidate lives in an asymptotically Poincaré-invariant region and possesses total four-momentum P^a , define its invariant mass by

$$m^2 c^2 := P_a P^a \quad (25)$$

(up to the chosen unit convention). This reverses the usual modelling order: mass is an observable/classifier of a stable solution, not a primitive variable used to create the solution.

Research criterion 10.2 (Ontic mass test). The proposal earns the statement “mass is emergent” only if localized stable solutions exist before assigning them a mass parameter and if their geometric influence correlates quantitatively with the invariant extracted from (25). Merely replacing a source parameter M by another fitted scalar does not satisfy this criterion.

10.3 Generations as branches, never as an input integer

Definition 10.3 (Generation candidate). Given a fixed set of asymptotic gauge charges and spin representation data, a *generation candidate* is a distinct stable connected component or isolated branch of the moduli space of matter candidates that cannot be continuously deformed into another such branch without crossing instability, changing conserved topological data, or leaving the admissible solution space.

The observed number three is a future comparison target. If the mathematical classification yields one, two, four, infinitely many, or a continuum of otherwise equivalent stable branches, the proposed generation mechanism is falsified rather than repaired by hand.

11 Propagating relaxation modes and the radiation question

Let $z_\star$ be a background solution and linearize the full equations/closure about it. A propagating mode is a non-gauge solution of the linearized system with finite flux and a well-defined dispersion relation.

Definition 11.1 (Radiation candidate). A *radiation candidate* is a propagating perturbation that carries conserved flux to asymptotic regions and is not pure gauge. This definition makes no assertion about polarization, particle interpretation or quantization.

The heuristic possibility that an unresolved local mismatch “breaks loose” is therefore translated into a mathematical question: does the linearized/nonlinear closure possess propagating modes that transport the relevant gauge-invariant defect/energy away from a localized configuration?

Research criterion 11.2 (Photon/thermal-radiation gate). A mode may be compared with electromagnetic radiation only after the theory derives the correct massless spin-1 representation with the observed physical helicities, gauge redundancy and coupling structure. It may be called thermal radiation only after a consistent quantization/statistical theory produces the correct equilibrium spectrum and thermodynamic relations. A fundamental mode that is longitudinal in some internal variable is not thereby an observed longitudinal photon.

12 Dense phases, confinement and deconfinement: only after an internal gauge sector exists

12.1 Why the analogy is tempting but insufficient

A dense collection of stable localized solutions may increase the incompatibility constraints on their shared relational background and may undergo bifurcations to new collective phases. This is a legitimate nonlinear-systems question. It is not yet QCD.

Yang and Mills showed that localizing a non-Abelian internal symmetry introduces nonlinear gauge fields [6]. Wilson’s lattice formulation supplied gauge-invariant loop variables and a strong-coupling confinement mechanism [14]. Gross–Wilczek and Politzer established asymptotic freedom [15, 16]. Cabibbo and Parisi connected an exponentially growing hadronic spectrum with a distinct phase in which quarks are not confined [17]; Collins and Perry argued that sufficiently dense matter should enter a quark regime in an asymptotically free theory [18]; and Shuryak explicitly analyzed observables of a quark–gluon plasma [19]. These are hard benchmarks for any attempt to connect the present programme to QCD or quark–gluon plasma physics.

Research criterion 12.1 (QCD entry gate). The programme may enter a QCD comparison phase only if it has independently produced an internal non-Abelian gauge sector with gauge-invariant observables. The Lorentz/spin connection alone is not an $SU(3)$ colour connection.

Benchmark 12.2 (Confinement benchmark). An emergent strong-interaction sector must exhibit a gauge-invariant confinement criterion in the appropriate regime, such as a nonzero asymptotic string tension or Wilson-loop area-law behaviour, and simultaneously recover short-distance asymptotic freedom. A phenomenological statement that “objects are hard to separate” is insufficient.

Benchmark 12.3 (Dense/deconfined phase benchmark). At high temperature/density the model must exhibit the correct qualitative and quantitative loss/modification of confining behaviour, screening and equation-of-state structure before its collective phase may be compared with quark–gluon plasma physics.

13 Fusion and restructuring of stable configurations

If stable localized solutions exist, collisions become a scattering problem in the same closure theory. “Fusion” is then not defined by visual merging but by a process

$$\mathcal{S}_\alpha + \mathcal{S}_\beta \longrightarrow \mathcal{S}_\gamma + \text{radiative modes} \quad (26)$$

that respects all exactly conserved charges and is dynamically allowed above a calculable threshold. Whether the final state is more tightly bound, fragments, or enters a dense collective phase is a result of the nonlinear equations, not a rule supplied externally.

14 The two-sided smoke test

The core hypothesis should be attacked first in a deliberately minimal model with two distinct boundary perturbations but one internal law.

14.1 State-side perturbation

Start from a neutral synchronization solution and perturb only relational/core state data at the boundary or initial surface. Do not input a velocity formula. Let the solution determine the resulting relative group element and background readjustment. The terminal flat-space comparison is special relativity.

14.2 Geometry-side perturbation

In an initial smoke test, one may impose a small geometric boundary perturbation to test reciprocity. The local core itself receives no direct force impulse. The state response must arise through the same closure law. This is only a reciprocity test; the mature model must replace the imposed background perturbation by one generated self-consistently by other cores.

14.3 Many-body closure test

Replace the imposed geometric perturbation by a localized cluster of many core configurations and solve all background variables simultaneously. A remote test core is then evolved in the resulting endogenous geometry. This is the first test that addresses the intended gravitational mechanism rather than an externally programmed curved background.

Research criterion 14.1 (One-law criterion). All three tests must use the same $\mathfrak{C}$ (or same action, if a variational closure is derived). Changing only boundary/initial conditions is permitted. Retuning the fundamental law between the kinematic and geometric tests invalidates the unification claim.

15 Research programme and dependency DAG

15.1 Stage 0: finish the self-contained formal spine

Before new physics-oriented formalization begins, the refactored project should satisfy a zero-legacy acceptance condition: the new spine builds with both historical experiment trees unavailable. Any historically required theorem must be rewritten natively in the new project. This is an engineering prerequisite, not part of the physical hypothesis.

15.2 Stage 1: complete the mathematical spin-geometry bridge

Goal: obtain a genuine Lorentzian spin-geometric substrate without dynamics.

- 1.1 Equip the native spin cover and Lorentz group with the required topology and topological-group structures.
- 1.2 Prove continuity, kernel discreteness and local continuous sections of the spin cover.
- 1.3 Construct the oriented/time-oriented Lorentz frame bundle and the global spin lift, with the correct w_1/w_2 obstruction logic and no conflation of manifold orientation with time orientation [9, 10, 11].
- 1.4 Construct associated Minkowski and spinor bundles.
- 1.5 Introduce/classify solder forms and spin connections without selecting dynamics.

Stop condition: the geometric objects and their dependencies are proved, but no Einstein or matter equation is assumed.

15.3 Stage 2: identify the minimal synchronization variables

Goal: determine whether the full spin/bundle structure canonically supplies a smaller comparison variable than an arbitrary pair (e, ω) .

- 2.1 Classify gauge-invariant local and loop comparison data.
- 2.2 Determine whether a scalar “imbalance” is canonical or merely a one-parameter boost reduction. Do not promote rapidity or a null-coordinate ratio to a fundamental field unless uniqueness is proved.
- 2.3 Separate curvature-type, torsion-type and purely gauge mismatch.
- 2.4 Define the neutral synchronization class invariantly.

Failure signal: if the proposed synchronization variable depends on arbitrary coordinates/frame choices, discard it.

15.4 Stage 3: finite many-fibre prototype

Goal: prove or refute the existence of a reciprocal self-consistency rule in a controlled discrete system.

- 3.1 Build a graph of identical local cores and native spin transports.
- 3.2 Establish gauge actions, loop holonomies and exact neutral/flatness criteria.
- 3.3 Search a constrained class of local gauge-invariant closure functionals/operators; record all failures.
- 3.4 Demand two-sided regular response and collective nonseparability.
- 3.5 Test whether nontrivial loop structure can be generated by clusters without changing the local fibre core.

Stop condition: no continuum or GR claims until a nontrivial reciprocal discrete closure survives falsification.

15.5 Stage 4: continuum limit and closure classification

Goal: pass from edge transports to (e, ω) or a more fundamental continuum variable.

- 4.1 Prove a controlled continuum limit, identifying which discrete observables converge to holonomy, curvature and/or torsion.
- 4.2 Classify local gauge-covariant low-derivative closure operators compatible with the inherited symmetries.
- 4.3 Determine whether the closure is variational; if so, derive the action rather than choosing Einstein–Hilbert/Palatini/Yang–Mills by target matching.
- 4.4 Audit all dimensionful scales.

15.6 Stage 5: flat-sector falsification against SRT

Goal: recover special relativity as a terminal theorem of the flat sector.

- 5.1 Set collective curvature/torsion to the derived neutral values, not by coordinate convention alone.
- 5.2 Apply a relational state perturbation.
- 5.3 Derive the relative Lorentz transformation from the native spin cover.
- 5.4 Compare only after derivation with rapidity/velocity, Doppler and simultaneity formulae.

15.7 Stage 6: self-generated weak geometry

Goal: determine whether many-core configurations generate a viable long-range geometric response.

- 6.1 Construct a localized many-core solution with no primitive mass source.
- 6.2 Solve the background variables self-consistently.
- 6.3 Insert a remote test core with no direct gravitational force term.
- 6.4 Derive its effective motion.

- 6.5 Compare with Newtonian free fall, gravitational redshift and weak-field GR.
- 6.6 Determine the effective field equation and compare with Einstein/Einstein–Cartan structures only after it is obtained.

Critical failure: if the correct response appears only after inserting a Schwarzschild lapse, Newton potential or Einstein tensor relation, the derived-gravity claim has failed.

15.8 Stage 7: stable localized solutions and mass

Goal: search for matter candidates only after the background closure exists.

- 7.1 Classify finite-energy localized solutions/moduli.
- 7.2 Prove stability or identify decay channels.
- 7.3 Derive asymptotic conserved charges and Poincaré invariants.
- 7.4 Define mass by (25) and test whether the same invariant controls long-range geometric influence.
- 7.5 Only then classify possible repeated stable branches and test the generation question.

15.9 Stage 8: propagating modes and thermodynamics

Goal: determine whether mismatch relaxation produces radiation candidates.

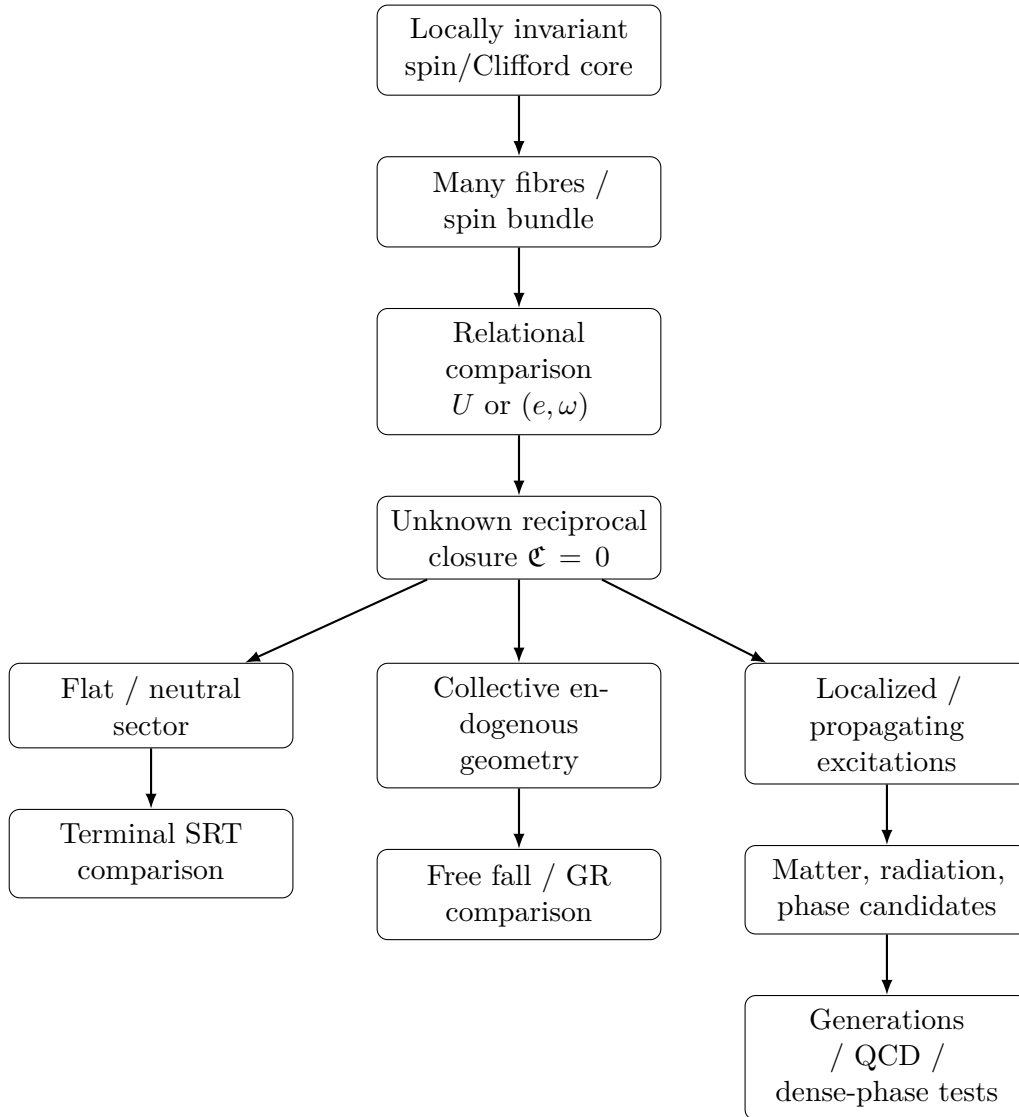
- 8.1 Linearize around vacuum and bound backgrounds.
- 8.2 Classify gauge versus physical modes, dispersion and polarization.
- 8.3 Quantize only if a consistent Hamiltonian/symplectic structure exists.
- 8.4 Test whether any mode reproduces electromagnetic observables.
- 8.5 Derive statistical mechanics before making a heat-radiation identification.

15.10 Stage 9: high-density and strong-interaction tests

Goal: enter QCD-like questions only if an independent internal non-Abelian sector has emerged.

- 9.1 Identify the internal gauge group and gauge-invariant observables.
- 9.2 Test confinement/string tension and short-distance running.
- 9.3 Test dense/high-temperature phase structure and deconfinement.
- 9.4 Analyze scattering, restructuring and possible fusion of stable states.

15.11 Dependency diagram



16 Failure modes and hard falsifiers

Falsification criterion 16.1 (No nontrivial closure). If every gauge-covariant reciprocal closure compatible with the local core is either trivial/pure gauge or requires importing the target dynamical equations, SCH fails at its central step.

Falsification criterion 16.2 (Two-law failure). If kinematic response and geometric/free-fall response require irreducibly independent fundamental laws rather than two boundary realizations of one closure, the proposed common mechanism fails.

Falsification criterion 16.3 (Local invariance failure). If fitting observations requires changing the local Minkowski/spin core from fibre to fibre rather than changing relational geometry, the stated hypothesis fails in its present form.

Falsification criterion 16.4 (Scale laundering). If masses or gravitational couplings appear only because a dimensionful target value was inserted under a renamed parameter or unit convention, the emergence claim fails.

Falsification criterion 16.5 (Matter failure). If the closure has no stable localized finite-energy solutions, the proposed matter branch fails even if the geometric branch survives.

Falsification criterion 16.6 (Generation failure). If stable-state classification does not naturally produce the observed generation pattern, the generation extension fails; the geometry hypothesis need not fail with it.

Falsification criterion 16.7 (Strong-sector failure). If no appropriate internal non-Abelian gauge sector emerges, no claim about confinement or QGP follows from SCH. If such a sector emerges but fails asymptotic-freedom/confinement benchmarks, the QCD extension is falsified.

17 Relationship to established primary literature

The purpose of this section is to delimit novelty rather than accumulate analogies.

Cartan (1923). Affine connections, curvature and torsion are established geometric structures [2]. SCH does not claim their invention; it asks what selects their physical configuration from the deeper core.

Fock and Weyl (1929). Spinors in gravitational/local-frame settings and the need for local frame/connection data are established [3, 4]. SCH asks whether the relevant coupling structure can be derived from the project's reconstructed dependency chain rather than appended as an external gravitational formalism.

Ambrose–Singer (1953). Curvature controls holonomy [5]. Therefore loop mismatch is a mathematically legitimate probe of non-flat comparison structure. This theorem does not supply a self-regulation law.

Yang–Mills (1954). Local non-Abelian symmetry leads to nonlinear gauge fields [6]. This is a benchmark for any later internal gauge sector, not a reason to identify spin holonomy with colour.

Kibble (1961). Localizing the inhomogeneous Lorentz group introduces vierbein and connection variables and provides a gauge-gravity action with spin-dependent torsion effects [7]. This is especially close structurally and therefore an essential control: a purported derivation that merely reconstructs Kibble after assuming his action has not established SCH.

Milnor and Geroch (1963–1970). Global spin structures have topological existence conditions and are additional structure beyond a Lorentz metric [9, 10, 11]. These conditions must be formalized before a global synchronization programme can legitimately speak of spinor fields on arbitrary manifolds.

Einstein and Schwarzschild (1916). General relativity and its exact spherically symmetric exterior solution provide macroscopic geometry benchmarks [12, 13]; they are explicitly forbidden as definitions of the new closure.

Skyrme (1961). Finite-energy particle-like nonlinear field configurations provide a primary precedent for emergent localized objects [8]. SCH requires its own stability mechanism and observables.

Wilson, Gross–Wilczek, Politzer, Cabibbo–Parisi, Collins–Perry, Shuryak. Gauge-invariant confinement, asymptotic freedom, quark liberation/deconfinement and dense/hot quark matter are stringent strong-interaction benchmarks [14, 15, 16, 17, 18, 19]. They belong only to a late branch of the programme.

18 Conclusion: the hypothesis in one statement

The research hypothesis can be stated compactly as follows.

Hypothesis 18.1 (Condensed form). There exists a deeper locally invariant spin/Clifford core whose many-fibre relational comparison data possess a gauge-covariant self-consistency law. The local core remains Minkowskian, while the relational background is an endogenous variable of that same law. State-side and geometry-side perturbations are two boundary realizations of one reciprocal closure. In suitable sectors, the flat response should reproduce special-relativistic kinematics; collective non-flat solutions should reproduce gravitational geometry and free fall; stable localized and propagating solutions may then be tested as matter and radiation candidates. Mass, generations and strong-interaction phases are not inputs but later invariants/classification outcomes, and every dimensionful scale must have explicit provenance.

The decisive mathematical unknown is therefore not “what is curvature?” but

What is the minimal native closure $\mathfrak{C}$ that makes the relational background self-consistent with many invariant cores?

If no such nontrivial closure exists, the programme should terminate at that point. If one exists, the subsequent SRT, gravity, matter, radiation and QCD comparisons are independent falsification stages, not guarantees. This ordering is essential: the target phenomena are known, but the defining equations that produce them must not be inserted backwards into the premises.

References

- [1] T. Tao, *Describe the results accurately*, in Advice on Writing Papers, What's New (2007). <https://terrytao.wordpress.com/advice-on-writing-papers/describe-the-results-accurately/>.
- [2] E. Cartan, *Sur les variétés à connexion affine et la théorie de la relativité généralisée (première partie)*, Annales scientifiques de l'École Normale Supérieure, 40 (1923), 325–412. DOI: [10.24033/asens.751](https://doi.org/10.24033/asens.751).
- [3] V. Fock, *Geometrisierung der Diracschen Theorie des Elektrons*, Zeitschrift für Physik 57 (1929), 261–277. DOI: [10.1007/BF01339714](https://doi.org/10.1007/BF01339714).
- [4] H. Weyl, *Gravitation and the Electron*, Proceedings of the National Academy of Sciences 15(4) (1929), 323–334. DOI: [10.1073/pnas.15.4.323](https://doi.org/10.1073/pnas.15.4.323).
- [5] W. Ambrose and I. M. Singer, *A Theorem on Holonomy*, Transactions of the American Mathematical Society 75(3) (1953), 428–443. DOI: [10.1090/S0002-9947-1953-0063739-1](https://doi.org/10.1090/S0002-9947-1953-0063739-1).
- [6] C. N. Yang and R. L. Mills, *Conservation of Isotopic Spin and Isotopic Gauge Invariance*, Physical Review 96 (1954), 191–195. DOI: [10.1103/PhysRev.96.191](https://doi.org/10.1103/PhysRev.96.191).
- [7] T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field*, Journal of Mathematical Physics 2(2) (1961), 212–221. DOI: [10.1063/1.1703702](https://doi.org/10.1063/1.1703702).
- [8] T. H. R. Skyrme, *A Non-Linear Field Theory*, Proceedings of the Royal Society A 260 (1961), 127–138. DOI: [10.1098/rspa.1961.0018](https://doi.org/10.1098/rspa.1961.0018).
- [9] J. Milnor, *Spin Structures on Manifolds*, L'Enseignement Mathématique 9 (1963), 198–203. Persistent DOI record: [10.5169/seals-38784](https://doi.org/10.5169/seals-38784).
- [10] R. Geroch, *Spinor Structure of Space-Times in General Relativity. I*, Journal of Mathematical Physics 9 (1968), 1739–1744. DOI: [10.1063/1.1664507](https://doi.org/10.1063/1.1664507).
- [11] R. Geroch, *Spinor Structure of Space-Times in General Relativity. II*, Journal of Mathematical Physics 11 (1970), 343–348. DOI: [10.1063/1.1665067](https://doi.org/10.1063/1.1665067).
- [12] A. Einstein, *Die Grundlage der allgemeinen Relativitätstheorie*, Annalen der Physik 354(7) (1916), 769–822. DOI: [10.1002/andp.19163540702](https://doi.org/10.1002/andp.19163540702).
- [13] K. Schwarzschild, *Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie*, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (1916), 189–196. Primary scan (SAO/NASA ADS): <https://adsabs.harvard.edu/pdf/1916SPAW.....189S>.
- [14] K. G. Wilson, *Confinement of Quarks*, Physical Review D 10 (1974), 2445–2459. DOI: [10.1103/PhysRevD.10.2445](https://doi.org/10.1103/PhysRevD.10.2445).
- [15] D. J. Gross and F. Wilczek, *Ultraviolet Behavior of Non-Abelian Gauge Theories*, Physical Review Letters 30 (1973), 1343–1346. DOI: [10.1103/PhysRevLett.30.1343](https://doi.org/10.1103/PhysRevLett.30.1343).
- [16] H. D. Politzer, *Reliable Perturbative Results for Strong Interactions?*, Physical Review Letters 30 (1973), 1346–1349. DOI: [10.1103/PhysRevLett.30.1346](https://doi.org/10.1103/PhysRevLett.30.1346).
- [17] N. Cabibbo and G. Parisi, *Exponential Hadronic Spectrum and Quark Liberation*, Physics Letters B 59(1) (1975), 67–69. DOI: [10.1016/0370-2693\(75\)90158-6](https://doi.org/10.1016/0370-2693(75)90158-6).

- [18] J. C. Collins and M. J. Perry, *Superdense Matter: Neutrons or Asymptotically Free Quarks?*, Physical Review Letters 34 (1975), 1353–1356. DOI: [10.1103/PhysRevLett.34.1353](https://doi.org/10.1103/PhysRevLett.34.1353).
- [19] E. V. Shuryak, *Quark-Gluon Plasma and Hadronic Production of Leptons, Photons and Psions*, Physics Letters B 78(1) (1978), 150–153. DOI: [10.1016/0370-2693\(78\)90370-2](https://doi.org/10.1016/0370-2693(78)90370-2).